Warnakulasooriya Umesh Ashen Lowe

Civil & Transportation Engineer | PhD researcher

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Professional Profile

Transport researcher at UCD with strong background in sustainable mobility, travel behaviour modelling, and infrastructure planning. Currently contributing to the NexSys energy research programme, specializing in data-driven travel behaviour modelling. Proven ability to communicate complex research data into actionable engineering insights for sustainable transport planning. Seeking to apply advanced research skills and modelling expertise within a leading Irish transport consultancy.

Skills

- **Transport Modelling and Design** : PTV Vissim | Visum| Civil 3D |QGIS | AutoCAD| MATSim | NetLogo |ArcGIS
- **Data Science** : Basic skills in C++, MATLAB, Mathematica, , SQL, | Experienced with R, Python, C++, MS Project, SAP 2000, Microsoft Office Excel, Word, and PowerPoint
- **Professional Skills :** Technical report writing | Mathematical Modelling | Multidisciplinary Approach | Stakeholder engagement
- **Soft Skills** : Analytical Thinking| Coordination Skills |Team Working| Communication skills| Adaptability

Career Summary

PhD Researcher, University College Dublin, Ireland 01/2023 – present

NexSys National Energy Research Programme · Thesis: Investigating potential ways to improve effectiveness of travel behaviour interventions aimed at reducing car use for short distance trips

- Developed an agent-based simulation model in NetLogo to evaluate the effectiveness of policy interventions (fuel taxation, pricing, active travel incentives) on modal shift and CO_2 emissions across Dublin — model published on the CoMSES Computational Model Library (2026).
- Conducted quantitative and qualitative analysis of transport policy data using R and Python to support Ireland's net-zero energy transition goals within NexSys.
- Collaborated with a multi-disciplinary team of economists, behavioural scientists and engineers across the NexSys programme, aligning transport outputs with national energy modelling frameworks
- Published peer-reviewed papers across international journals spanning travel behaviour, transport policy, and sustainable mobility.

Teaching Assistant, University College Dublin, Ireland 01/2023 – present

- Delivered weekly lab sessions in Transportation Engineering for 20+ students per semester covering Python, GIS, PTV Visum and Vissim.
- Invigilated examinations for 1,000+ students per semester; demonstrated professionalism, organisation and reliability at scale

Researcher, SIIT, Thammasat University, Thailand 09/2022 – 11/2022

- Applied machine learning techniques to identify factors influencing carpool adoption in Thailand; produced technical reports on public transport policy support during COVID-19.

Teaching Assistant, SIIT, Thammasat University, Thailand 08/2020 – 08/2020

- Supervised projects for undergraduate senior research project in questionnaire survey and using R for data analysis.
- Delivered weekly lab sessions 40+ students related to the soil mechanics, Mathematica based computer graphics and exam invigilation, both online and onsite in each semester.

Research Assistant, University of Peradeniya, Sri Lanka 02/2020 – 08/2020

- Collected data collaborating with local transport authorities and reported for public transport development research project. Develop a model for bus transport, and a transport network in GIS and transform findings into reports.

Instructor, University of Peradeniya, Sri Lanka 03/2019 – 01/2020

- Delivered academic lab sessions for 150 + students in the soil mechanics, survey, material and fluid mechanics laboratory.
- Delivered structural design classes using Euro codes for 150+ students.

Civil Engineer (Intern), Sierra Construction Ltd, Sri Lanka 11/2017 – 01/2018

- Assisted consultants with inspections. Guide labourers and report the progress to the site Engineer.

Civil Engineer (Intern), Provincial Engineering Services Department, Sri Lanka 11/2016 – 01/2017

- Work with the consultants' team. Inspect the contractor's work and report it to the consulting Engineer.

Education

PhD (Civil Engineering), University College Dublin, Ireland Expected completion 2026

MSc (Engineering and Technology), SIIT, Thammasat University, Thailand 2022

BSc (Specialised in Civil Engineering), University of Peradeniya, Sri Lanka 2019

Continuing professional development: AI in Urban mobility |AI Fluency Framework & Foundations |Transportation Planning and Safety |Research integrity| Safety induction |SQL for data science |Data analysis with R |Data foundation

Research experience

- **Projects:** Next Generation Energy Systems (NexSys), a multidisciplinary energy research program (2023-2026) | Assessment of Transport Policies to Tackle Global Pandemics for Sustainable Mobility in ASEAN Countries (2021-2022)| Enhancing bus-based public transport system in central province Sri Lanka (2020-2021)
- **Publications:** European Transport Research Review-2024| Case Studies on Transport Policy- 2023|Journal of the Eastern Asia Society for Transportation Studies-2022|Proceedings of the 2021 9th International Conference on Information Technology: IoT and Smart City -2021 | ICSBE 2018: Proceedings of the 9th International Conference on Sustainable Built Environment-2020

Professional Affiliations

- **Volunteer organising committee:** International Network Optimisation Conference (INOC) 2024 | SCICOM conference 2023| Transport Research symposium 2023 by Transport Research Hub at UCD
- **STEM demonstrator volunteer:** BT Young Scientist and technology exhibition 2025/2026| *I'm a scientist* chat program |UCD STEM Adventure program | Math Teacher secondary-level students who lacked educational facilities in Sri Lanka
- **Co-curricular Activities:** Waterford volleyball club |Dunmore badminton club | University baseball team | School baseball, cricket, table tennis and naval cadet member
- **Memberships:** Transport Research Hub at UCD | Sri Lankan Society at UCD | Thammasat International Club| Civil Engineering Society at UOP | Associate member Institute of Engineers Sri Lanka

References

Available on request